

How do different tokenizers perform on downstream tasks in scriptio continua languages?: A case study in Japanese

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Inspire the Next

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Summary

- We investigate the effect of tokenizers on the downstream tasks in scriptio continua languages using Japanese as a case study.
- **Findings**
 - We should use a morphological analyzer for Japanese.
 - Each task seems to have its own optimal morphological analyzer(s).
 - It is better to use either BPE or Unigram as a subword tokenizer rather than WordPiece.

Background

- Tokenizers are tools to split sentences into word or subword units.
- Different tokenizers have **different tokenization outputs**, and it is widely acknowledged that **tokenization affects downstream performances of PLMs**.
- Research on tokenizers of scriptio continua languages is still understudied.

Tokenizers for Scriptio Continua Languages

- **Scriptio continua languages** have no explicit spaces between words such as Japanese, Chinese and Thai.

Examples of Scriptio Continua Languages

- ✓ (Original text: "I am doing research on morphological analyzers.")
- ✓ In Japanese: "私は形態素解析器の研究をしています。"
- ✓ In Chinese: "我正在研究形态分析器。"
- ✓ In Thai: "ฉันกำลังค้นคว้าเกี่ยวกับเครื่องวิเคราะห์หลักฐานวิทยา"

- Such tokenizers often consist of a pair of a **morphological analyzer** and a **subword tokenizer**.
- There are various morphological analyzers and subword tokenizers.

Tokenizer Composition



Figure 1: Examples of morphological analyzers and subword tokenizers.

The Tokenization Procedure of English (ENG) and Japanese (JPN)

Original text:

(ENG) I am doing research on morphological analyzers.
(JPN) 私は形態素解析器の研究をしています。

Step 1: Morphological analysis (Splitting into "word-level" semantic units)

(ENG) I_am_doing_research_on_morphological_analyzers.
(JPN) 私_は_形態素_解析_器_の_研究_を_し_て_い_ます_。

Step 2: Subword tokenization

(ENG) I / am / doing / research / on / morphological / analyze / ##rs / .
(JPN) 私 / は / 形態 / ##素 / 解析 / 器 / の / 研究 / を / し / て / い / ます / 。

- Publicly available pretrained language models use **different tokenizers**.
 - ✓ c1-tohoku/bert-base-japanese: **MeCab** and **WordPiece**
 - ✓ nlp-waseda/roberta-base-japanese: **Juman++** and **Unigram**

Research Questions

RQ1: Should we use a morphological analyzer?

RQ2: Do different morphological analyzers perform differently on downstream tasks?

RQ3: Do different subword tokenizers perform differently on downstream tasks?

Experimental Setup

- **Tokenizers:** We compare a total of 12 tokenizers (4 morphological analyzers and 3 subword tokenizers), as introduced in Figure 1.
- **Models:** We use the base configuration of BERT. For each tokenizer, we pretrain BERT on Wikipedia and CC-100 datasets.
- **Benchmarks:** JGLUE, NER and Universal Dependencies (UD).

Results and Answers for RQs

- **RQ1:** "Nothing (no morphological analyzer)" is the worst result in almost tasks and subword tokenizers. [📄 Table 1]
 - ➡ **We should use a morphological analyzer in Japanese.**
 - Morphological analyzers can provide explicit semantic boundaries of and input text, which should help a model to capture the semantic and syntactic information more easily.

- **RQ2:** There are significant performance difference between different morphological analyzers with the same subword tokenizers in some tasks, e.g., JSTS, NER, and UD. [📄 statistical test result in Table 1]
 - ➡ **Different morphological analyzers could perform differently on downstream tasks.**
 - Especially, there are task-specific trends for an effective morphological analyzer(s). [📄 Table 2]
 - Vaporetto perform well for JSTS, and Juman++ is effective for NER.
- **RQ3:** There are significant performance difference between different subword tokenizers with the same morphological analyzers in some tasks, e.g., JSTS and JCQA. [📄 statistical test result in Table 1]
 - ➡ **Different subword tokenizers could perform differently on downstream tasks.**
 - **Addition:** We analyze which differences in subword tokenizers produce downstream performance differences.
 - ● and ▲ are plotted in the upper-left corner. [📄 Figure 2]
 - WordPiece has a different vocabulary composition than BPE and Unigram and its performance difference is far larger than that between BPE and Unigram.
 - WordPiece uses a greedy longest-match-first strategy in tokenizing a word, both BPE and Unigram use a more sophisticated approach.

Tokenizer		MARC-ja	JSTS	JNLI	JSQuAD	JCQA	NER	UD	Avg.
subword	Morphological	Accuracy	Spearman	Accuracy	F1	Accuracy	F1	LAS	
BPE	M MeCab	95.4	84.2	88.0	90.1	74.1	83.7	93.6	87.0
	J Juman++	95.5	84.6	87.6	90.1	73.8	85.1	93.6	87.2
	S Sudachi	95.5	84.2	88.2	90.2	74.2	83.5	93.8	87.1
	V Vaporetto	95.6	84.8	87.5	89.9	74.2	84.1	93.7	87.1
	Nothing	95.4	82.8	87.2	88.7	72.8	62.9	93.4	83.3
WordPiece	MeCab	95.5	82.4	87.5	89.2	69.8	84.0	93.6	86.0
	Juman++	95.3	83.3	87.7	89.8	71.1	84.7	93.6	86.5
	Sudachi	95.3	83.7	87.2	89.6	70.0	82.4	94.0	86.0
	Vaporetto	95.3	83.6	88.0	89.7	71.0	84.0	93.8	86.5
	Nothing	85.5	N/A	55.3	10.1	20.0	0.0	63.8	33.5
Unigram	MeCab	95.4	84.6	88.3	89.5	74.5	83.1	93.4	87.0
	Juman++	95.4	84.3	87.8	89.9	74.9	84.1	93.4	87.1
	Sudachi	95.6	84.8	88.4	89.9	74.5	83.0	93.7	87.1
	Vaporetto	95.5	84.6	87.9	89.9	74.3	84.1	93.7	87.1
	Nothing	95.4	83.9	97.7	89.3	74.6	76.9	93.2	85.9

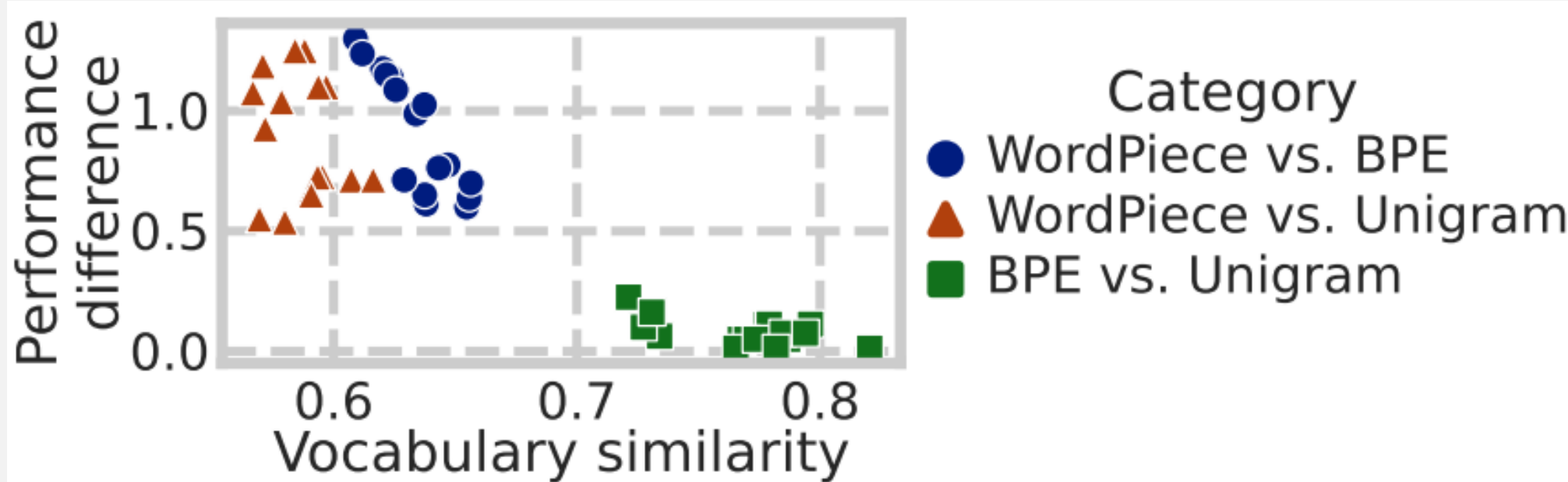
Statistical test results: Kruskal-Wallis test. ✓ if $p < .05$ otherwise ✗.

RQ2: (B, W, U)	(✗, ✗, ✗)	(✓, ✓, ✗)	(✓, ✗, ✗)	(✗, ✗, ✗)	(✗, ✓, ✗)	(✓, ✓, ✗)	(✓, ✓, ✓)
RQ3: (M, J, S, V)	(✗, ✗, ✗, ✗)	(✓, ✓, ✓, ✓)	(✗ ✗ ✗ ✗)	(✓ ✗ ✗ ✗)	(✓, ✓, ✓, ✓)	(✗, ✗, ✗, ✗)	(✗, ✗, ✓, ✗)

📄 Table 1: Results from seven tasks with standard deviations over five runs. JCQA stands for JCommosenseQA. Blue values denote the worst results among morphological analyzers with the same subword tokenizer. ✓ indicates that there is statistical significance, while ✗ denotes that there is no statistical significance.

	JSTS	JNLI	JCQA	NER	UD
BPE	(V > M)	-	-	(J > S)	(S > M)
	(V > S)	-	-		(S > J)
WordPiece		-	-		(S > M)
	(S > M)	-	-	(J > S)	(S > J)
	(V > M)	-	-		(V > M)
Unigram		-	-		(V > J)
	-	-	-	-	-

📄 Table 2: Combinations of morphological analyzers with statistical significance ($p < .05$, Steel–Dwass test). “-” indicates no statistical significance observed. “A > B” indicates that morphological analyzer “A” is significantly better than morphological analyzer “B”.



📄 Figure 2: Relationship between vocabulary similarity of subword tokenizers and their performance difference.